Code No: 138GY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
SENSORS AND TRANSDUCERS
(Common to EEE, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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JNTUH Used Papers

1.a) Distinguish between Direct comparison method and Indirect comparison methods of measurement.
b) Explain the following types of errors with respect to measurement systems:
i) Gross errors ii) Systematic errors iii) Random errors. [8+7]

2.a) What are Static characteristics and explain the following with respect to it:
i) Accuracy ii) Sensitivity iii) Reproducibility
iv) Drift v) Static error vi) Dead zone
b) Explain the time response of a first order system to a ramp input. [8+7]

3.a) List and explain the factors which influence the choice of transducers.
b) Define the term transducer and give the broad classification of it along with brief explanation of each. [8+7]

4.a) Explain the working of Resistive Potentiometers with necessary diagram and its characteristics.
b) Prove that the "gauge factor is equal to the summation of resistance change due to change of length, resistance due to change in area and resistance change due to Piezo resistive effect" with respect to Strain gauges. [5+10]

5.a) Explain the working of Linear variable Differential Transformer (LVDT).
b) Explain the working of resistance thermometer. [8+7]

6.a) Explain the working of Piezo – Electric transducers.
b) Two plates of parallel plate capacitive transducer are 32 mm apart and the space is filled with two different dielectric materials, one material is 1 cm thick with a dielectric constant of 5 and the 20 mm thick with a dielectric constant of 12. If the capacitive transducer were to be made up of a single dielectric material? What is the dielectric constant of that material? [8+7]

7. Name different mathematical model of transducer and explain in detail any one with suitable diagram. [15]

8. Write short notes on the following:
a) Hot wire anemometer b) Proving Ring. [7+8]

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